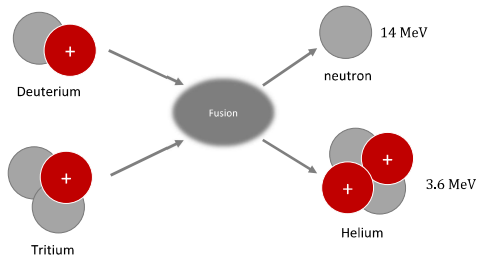
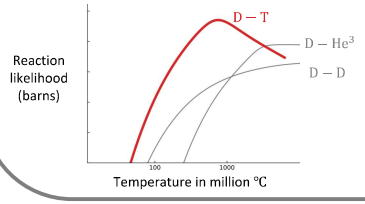
ARC

Affordable Robust Compact tokamak

Concept from:
Sorbom et al (2015), Fus. Eng. Design,
DOI: 10.1016/j.fusengdes.2015.07.008

What is Deuterium-Tritium fusion?



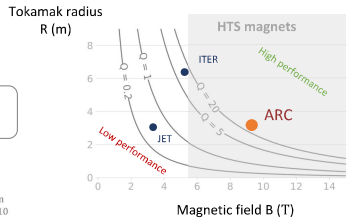


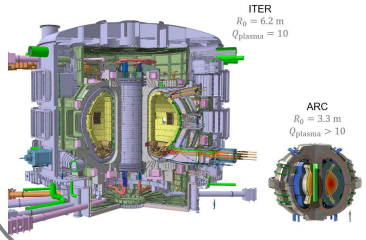
← Why do we need Deuterium and Tritium?

High Temperature Superconducting Magnet

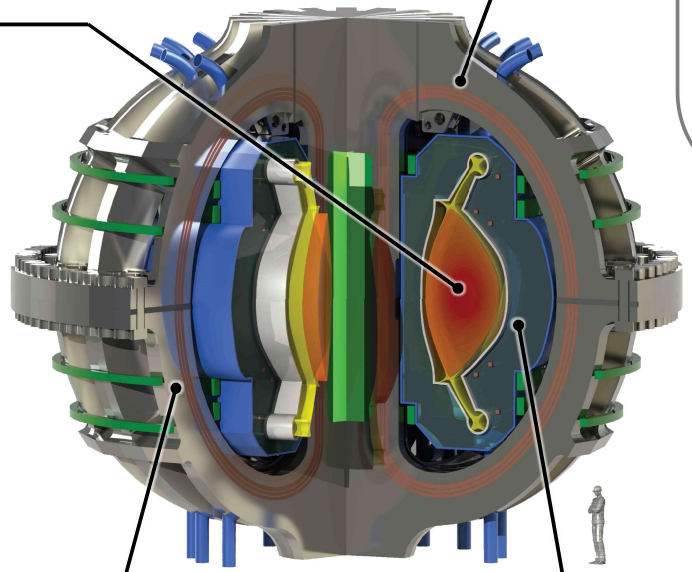
Increasing the magnetic field greatly improves fusion performance

$$Q_{\text{plasma}} = \frac{\text{fusion power}}{\text{heating power}} \propto R^3 B^5$$





Using HTS magnets allows tokamaks to be more compact and cheaper

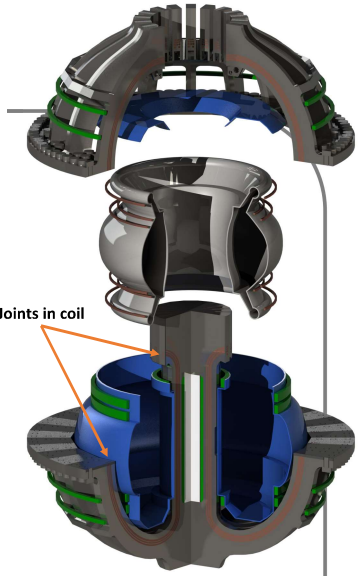


Major radius	3.3 m
TBR	1.1
Fusion power	525 MW
Multiplication factor Q	> 10
Magnetic field	9.2 T

Demountable magnets

Using **H**igh **T**emperature **S**uperconductors enables joints:

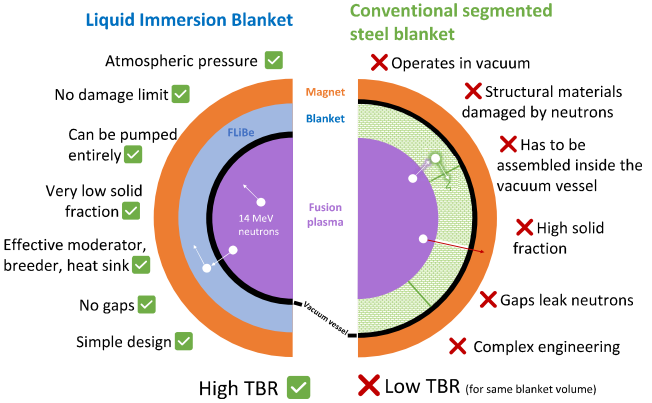
- ✅ Magnets can be mounted/demounted many times
- ✅ Easier maintenance from the top
- ✅ Vacuum vessel can be replaced and is no longer a lifetime component



ARC's assembly video

Liquid Immersion Blanket

Breeding tritium · Shield from neutrons · Extract heat



Why Lithium?

$$D + T \rightarrow He + n$$
$$Li + n \rightarrow T + He$$

Why Beryllium?

$$^9Be + n \rightarrow {}^8Be + n + n$$

$$\text{Tritium Breeding Ratio} = \frac{\text{Tritium produced}}{\text{Tritium consummed}}$$

Theoretical sufficiency → TBR = 1
Losses + start other plants → **TBR ≈ 1.1 − 1.2**